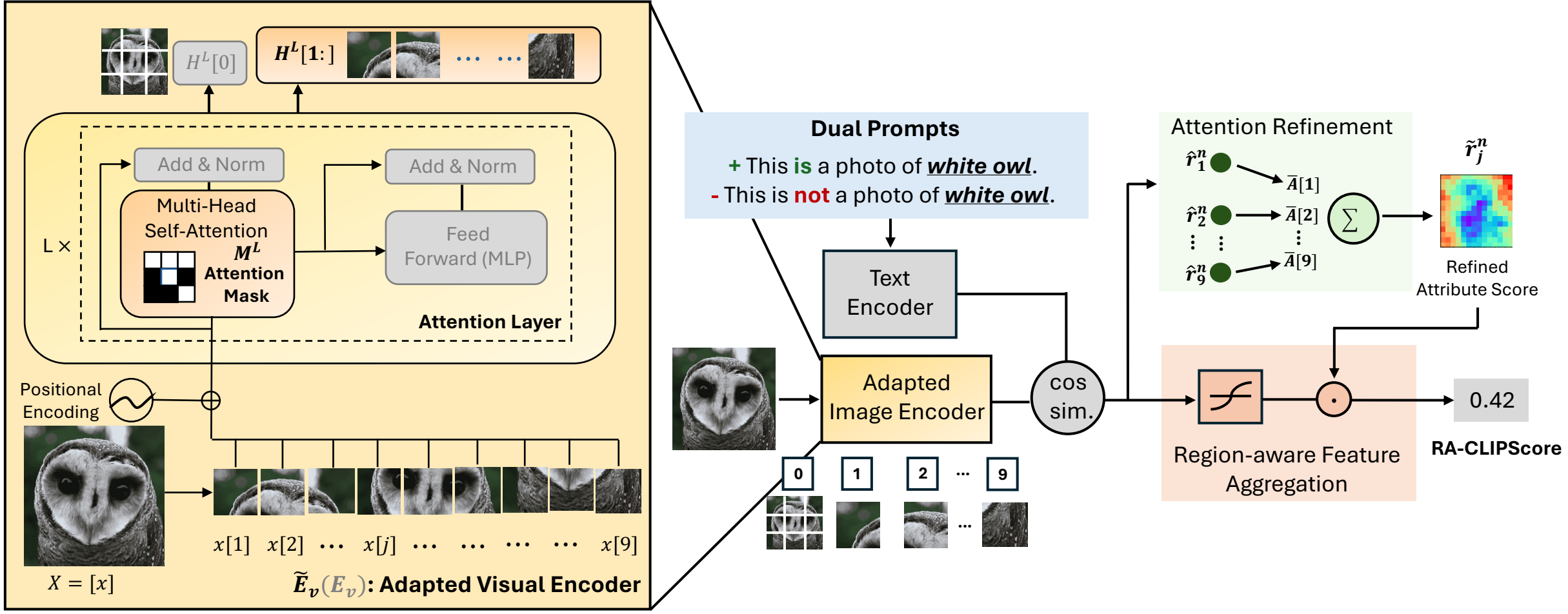

$H^L[0]$
$H^L[1:]$
Add & Norm
Multi-Head Self-Attention
M^L Attention Mask
Add & Norm
Feed Forward (MLP)
L ×
Attention Layer
Positional Encoding
$x[1]$ $x[2]$ ⋯ $x[j]$ ⋯ ⋯ ⋯ ⋯ $x[9]$
$X = [x]$
$\widetilde{E}_v(E_v)$: Adapted Visual Encoder
Dual Prompts
+ This is a photo of white owl.
- This is not a photo of white owl.
Text Encoder
Adapted Image Encoder
cos sim.
0
1
2
...
9
Attention Refinement
$\hat{r}_1^n$ $\hat{r}_2^n$ $\hat{r}_9^n$
$\bar{A}[1]$ $\bar{A}[2]$ $\bar{A}[9]$
$\sum$
$\tilde{r}_j^n$
Refined Attribute Score
Region-aware Feature Aggregation
0.42
RA-CLIPScore